

BOUNDARY STRATIFICATIONS OF HURWITZ SPACES

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ABSTRACT. Let $\mathcal{H}$ be a Hurwitz space that parametrises holomorphic maps to $\mathbb{P}^1$. Abramovich, Corti and Vistoli, building on work of Harris and Mumford, describe a compactification $\overline{\mathcal{H}}$ with a natural boundary stratification. We show that the irreducible strata of $\overline{\mathcal{H}}$ are in bijection with combinatorial objects called decorated trees (up to a suitable equivalence), and that containment of irreducible strata is given by edge contraction of decorated trees. This combinatorial description allows us to define a tropical Hurwitz space, refining a definition given by Cavalieri, Markwig and Ranganathan. We also discuss applications to complex dynamics.

1. INTRODUCTION

A *Hurwitz space* $\mathcal{H}$ parametrises holomorphic maps of compact Riemann surfaces (with certain points of the source and target surfaces marked) that satisfy prescribed branching conditions, up to pre- and post-composition with isomorphisms. These spaces are complex orbifolds [Ful69]. Harris and Mumford give a compactification by a space of *admissible covers*, which are maps between nodal Riemann surfaces [HM82]. The space of admissible covers is not normal, but its normalisation $\overline{\mathcal{H}}$ is smooth. Abramovich, Corti and Vistoli reinterpret $\overline{\mathcal{H}}$ as a space that parametrises *balanced twisted covers* [ACV03]. The boundary of $\overline{\mathcal{H}}$ is a normal crossings divisor, and this endows $\overline{\mathcal{H}}$ with a stratification by locally closed strata.

Cavalieri, Markwig and Ranganathan [CMR16] assign a combinatorial type called a *combinatorial admissible cover* to each point of $\overline{\mathcal{H}}$, building on work of Harris and Mumford. Points that share a given combinatorial type form a locally closed suborbifold that is a union of strata of the same dimension.

For a Hurwitz space $\mathcal{H}$ that parametrises maps to $\mathbb{P}^1$, the boundary of $\overline{\mathcal{H}}$ is actually a simple normal crossings divisor. In this case, we provide a more refined combinatorial type that describes the irreducible strata of $\overline{\mathcal{H}}$ exactly. The prescribed branching conditions that determine $\mathcal{H}$ are encoded in a *portrait* P (Definition 2.5). We define objects called *P -decorated trees* (Definition 4.1), and a suitable equivalence relation on P -decorated trees called *Hurwitz equivalence* (Definition 4.6), and we assign to each point of $\overline{\mathcal{H}}$ a Hurwitz equivalence class of P -decorated trees. We prove that two points of $\overline{\mathcal{H}}$ are in the same stratum if and only if they are assigned the same Hurwitz equivalence class of P -decorated trees.